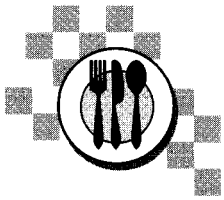
CHAPTER 15



Beverage Plant Sanitation

About This Chapter

In this chapter you will learn:

1. How sanitation is important to producers of nonalcoholic beverages, beer, wine, and distilled spirits
2. Why beverage plants are easier to clean than food-processing plants
3. Why it is essential to make sure that the raw ingredients used in beverages are pure
4. How to clean and sanitize different types of beverage plant
5. Why beverage producers need good-quality water and how they obtain it
6. Why sanitation needs to be strictest in the bottling area

Introduction

Soils found in beverage plants contain mostly sugar and dissolve in water easily, so beverage plants are easier to clean than other food-processing plants. Cleaning and control of microbes are more difficult in breweries and wineries than in plants that produce nonalcoholic beverages. Therefore, most of this chapter will concentrate on these two areas.

Focus on Yeasts

In beverage plants such as breweries, it is important to keep the useful microbes and remove those that cause spoilage and disease. Effective sanitation can keep microorganisms under control, but the plant will never be completely free of microorganisms. So if sanitation is poor, microorganisms soon contaminate beverages. Breweries are very similar to other manufacturing plants, although the most common

pathogens are not usually a problem, mainly because of the raw materials, processing techniques, and characteristics of the final product (acidity, high alcohol concentration, and carbon dioxide content). It is important to make sure that the raw materials are kept sanitary throughout processing because there is no good way to detoxify contaminated beverages once they are made.

SANITATION MANAGEMENT

Toilet facilities must be adequate, kept clean, and located close to the bottling area and other production areas. Employees must wash their hands after using toilet facilities. Drinking fountains should have guards so that a person's mouth or nose cannot touch the water outlet.

Employee Practices

As with other food operations, sanitation is a team job. In beverage plants, employees should clean as they go. Regular cleaning helps keep the plant tidy, reduces contamination, and shortens cleanup time at the end of the production shift or when switching production from manufacture of one product to another. Employees who operate equipment that fills bottles or cans often have time to pick up debris, hose down spills, etc.

Beverage plants need standards for employee work habits and a training program to make sure employees meet the standards. Supervisors need to use frequent updates, training, and educational materials to make sure that good work habits and sanitation are priorities. Employees should be told how, when, and where to clean so that soil and debris are removed before they can provide food for pests and microorganisms. Leaky equipment should be fixed immediately. If rodents, birds, insects, or molds are found, employees should get rid of them, if possible, and report the problem to management. Employees should store products properly to discourage pests and allow easy cleaning. Employees should keep doors and windows closed.

The following are sanitation rules for beverage plants:

1. All employees visiting toilet facilities must wash their hands before returning to work.
2. Employees may not use spilled ingredients or products; they must throw them away.
3. Employees must put waste in containers with tight-fitting lids.
4. Each employee must keep his or her work area clean and tidy.
5. Employees may not use tobacco, except in designated areas, away from food-handling areas.
6. Employees may not spit anywhere in the plant.
7. Managers should inspect clothing, lunchrooms, and lockers regularly to make sure they are clean.
8. Employees must always wear head covers.

Cleaning Methods

There are five standard steps for cleaning a beverage plant:

1. Rinse to remove large debris and soil that is not stuck on, to wet the area and to help the cleaning compound work properly.
2. Apply a cleaning compound (usually as foam) to wet and penetrate the soil.
3. Rinse to remove the dispersed soil and the cleaning compound.
4. Sanitize to destroy microorganisms.
5. Rinse again if using a quaternary ammonium sanitizer (especially if the concentration is >200 ppm) before the cleaned area touches any beverage products.

NONALCOHOLIC BEVERAGE PLANT SANITATION

This book cannot discuss sanitation of all types of nonalcoholic beverage plants. Ultra-high-temperature (UHT) aseptic packaging is becoming more common and will probably be the "wave of the future," but this technology is too complex to cover here. A technical publication about aseptic technology can provide more information about this method.

Proper hygiene in a beverage-processing facility requires sanitary water, steam, air, carbon dioxide, and packaging. Many beverage processors use various types of filter to make sure that liquids and gases are free of microorganisms and other contaminants.

Beverages such as soft drinks, bottled water, beer, and distilled spirits need to be made from pure water, so manufacturers often need to treat the water. Processors use flocculation, filtration (through a sand bed), chlorination, sterile filtration, reverse osmosis, activated carbon, and deionization. The type of treatment depends on how the water will be used.

Beverage producers condition water by removing particles and controlling microbes. Particles are usually removed by flocculation and sand filtration. Microbes are controlled by chlorination.

Treatment with activated carbon removes excess chlorine, trihalomethanes, and other compounds used to disinfect the water. But activated carbon sheds fine particles of carbon that can provide sites for microbes to grow. Also, carbon beds are difficult to disinfect. Therefore, the water is usually filtered again after leaving the carbon beds to reduce the load of microorganisms and particles.

Resin beds may be used to deionize water, but microbes can grow in the beds and beads of resin can get into the water. Again, the water is usually filtered after treatment in a resin bed. Sterile filtration does not use chemicals, is easy to use, and uses very little energy.

Steam is often used during production, but it can be a source of contamination. Steam is normally generated in carbon steel boilers, which rust easily. When a boiler operates continually, it develops a fine deposit of rust on the inside, which protects against further corrosion. When the boiler operates intermittently, fresh air gets inside and oxidizes the iron to iron oxides or rust. Rust forms continuously and flakes off into the steam.

Particles of rust from the boiler can damage equipment surfaces, block steam valves and filter pores, and stain equipment surfaces. Also, rust makes heat exchangers less efficient at heat transfer. To prevent this problem, processors often use culinary steam that is filtered using porous stainless-steel filters. The equipment has two parallel sets of filters so that one set can be cleaned while the other set is in use without interrupting the supply.

Many bottlers have installed cleaning-in-place (CIP) equipment to clean tanks, processing lines, and filters. Most bottlers that make several flavors of soft drink like CIP because it prevents the flavor "carry-over" from one drink to the next (especially from root beer).

One approach to cleaning beverage plants is known as TACT (time, action, concentration, and temperature). Using this method, it is possible to clean properly using different combinations of factors. For example, a 1% cleaning compound concentration at 43.5°C (110.3°F) may be the same as a 0.5% concentration at 60°C (140°F). However, it is important to remember that hotter water increases energy costs and condensation and cooks soil onto the surface being cleaned.

This section discusses cleaning in carbonated beverage plants, but other beverage plants use similar methods. Cleaning of floors, walls, and the bottling area are discussed under winery sanitation later in this chapter. Many cleaning methods in the beverage industry are similar to those used in dairy-processing plants (see Chap. 11).

Removing Tire Tracks

Tire tracks are difficult to remove. Alkaline cleaners work best. A mechanical scrubber helps remove the soil completely and more easily. Tire tracks are easier to remove and other soil will not be ground into the floor if the floor is cleaned every day.

Removing Soil from Conveyor Tracks

Soil on conveyor tracks is mostly spilled product, grease from bearings, filings from containers and tracks, and soap. A detergent track lubricant helps reduce contamination. Foam cleaning and a high-pressure rinse work well for cleaning conveyors.

Film Deposits

Films may form inside storage tanks, transfer lines, and filters. Thin films cause a dull surface, but more buildup looks bluish, and a thick film looks white. Residues from sugars are quite easy to remove, but aspartame and some gums make films that are difficult to remove. Processors may clean tanks manually, but circulation cleaning is more common. Chlorinated cleaning compounds work well for removing films.

Biofilms

Beverages and their ingredients have nutrients that allow microbes to grow and form biofilms. Biofilms can grow inside cooling towers, inside and outside warmers and

pasteurizers, and inside coolers. As for films, chlorinated alkaline cleaning compounds work best to remove biofilms. A quaternary ammonium sanitizer helps stop biofilms from forming.

Hot Sanitizing

Sanitation in beverage plants is different from that in other food facilities. During the past few years, hot sanitizing has been used more and more. Hot sanitizing works well for surfaces that come in contact with beverages, such as batch tanks, flow mix units, fillers, and coolers. Hot sanitizing is not economical because it uses so much energy and does not remove bacteria well, but it does penetrate well. Heat is good at penetrating equipment and destroying microorganisms behind gaskets or in tiny crevices.

Hot sanitizing is not the same as sterilization. Hot sanitizing raises the surface temperature to 85°C (185°F) for 15 minutes; sterilization needs 116°C (241°F) for 20 minutes. Sanitizing reduces the number of microbes to a safe level, but a few of the most resistant microorganisms (yeasts and spores) stay alive. Chemical sanitizers can kill as many microbes as hot sanitizing but work much more quickly.

Special cleaning compounds can be added during hot sanitizing to loosen and remove soils and biofilms. These compounds are designed to remove soil, condition the water, and be rinsed off during hot sanitizing. Soil and biofilm removal are essential for good sanitation.

BREWERY SANITATION

The environment in a brewery restricts growth of pathogens and spoilage microbes. Bacteria found in breweries can cause several problems, such as changing the pH of the wort (the wort is the liquid that is fermented to make beer), making it more acid or alkaline, adding acetic acid, preventing complete fermentation, and making "ropes" in the wort. Bacteria may also cause off-odors and hazes in the finished beer.

Control of Microbial Infection

Undesirable bacteria, yeasts, and molds grow rapidly in freshly cooled wort if it is contaminated. Therefore, workers must clean and sanitize brewery equipment that processes the wort. High-speed equipment such as fillers, cappers, casers, and keggers perform better if they are kept clean.

The best way to prevent beverage products from spoiling is to develop and maintain a comprehensive cleaning and sanitizing program. Sanitation personnel, a reliable sanitation consulting group, or a dependable cleaning compound and sanitizer supply firm can develop the program. Chapters 5 to 7 give more information on this topic. CIP (Chap. 7) is used more and more often for cleaning beverage equipment.

Fermentation facilities such as breweries need sterile air to make starter cultures and maintain sterile conditions inside storage tanks.

Cleaning Compounds

It is important to use the right cleaning compound for the type of soil. When spray cleaning, the cleaning compound should not make foam, because foam stops the solution from touching every part of the surface. The right cleaning compound will not allow beerstone (a deposit that can form on brewery equipment) to form. The cleaner should not attack metal and must be easily rinsed away so that the taste of the cleaner does not get into the beer. (See Chap. 5 for more information about cleaning compounds.)

Sanitizers

Chlorine or iodine are good sanitizers for fermenters, cold wort lines, and coolers. These sanitizers are added with the final rinse. Water can contain microorganisms; therefore, the final rinse should contain a sanitizer so that the final rinse does not leave bacteria or yeasts on the sterile surfaces of the equipment. (See Chap. 6 for more information about sanitizers.)

Heat Pasteurization

Producers of packaged beer use heat pasteurization to control microbes. Heat pasteurization is convenient, but it uses a lot of energy, which makes it expensive. Beverage producers have tried different methods because heat pasteurization:

- Has high energy costs
- Alters the flavor of beer
- Requires equipment that occupies space
- Makes working areas hot and humid

The alternative is cold pasteurization, using either:

- Chemicals, such as propyl gallate
- Millipore filtration and aseptic packaging or millipore filtration and other chemical treatments

New technology and safety information affect whether various chemical treatments are officially approved.

Aseptic Filling

Aseptic filling is different from pasteurization. Aseptic filling uses ultrafiltration to remove spoilage organisms from beer before packaging. Ultrafiltration occurs before packaging, so spoilage microorganisms can still get into the beer before it is sealed. The following suggestions help protect the product during aseptic filling:

- Filling rooms must be closed and have positive pressure of filtered air.

- Workers' clothes should always be clean.
- All involved should wash their hands with a sanitizing hand soap before entering the room.
- The conveyor belt should have a lubricant system that reduces microorganisms.
- The inside of the filler should be cleaned and sanitized daily, using recirculating CIP equipment.
- The outside of the filler, conveyors, connected equipment, floors, and walls must be foamed or gelled and sanitized every day. The detergent or sanitizer that is left on the surface after cleaning should prevent microbes from recontaminating the sanitized surfaces.
- Surfaces and air in the filling area should be monitored regularly to check for, bacteria, yeasts, and molds. A HACCP (Hazard Analysis Critical Control Point) program can include chemical and microbial monitoring to guarantee safe food production. The results should be checked to make sure levels of contaminants do not go too high and that levels do not slowly increase.

Bottle Cleaning

Centralized high-pressure, low-volume cleaning equipment has improved bottle cleaning (see Chap. 7). This method can remove soil from very hard-to-reach equipment such as conveyors, bottle fillers, cappers, and casers.

Sanitation in Storage Areas

Edible dry products—such as grain, sugar and other dry products—need proper storage. Workers should clean screw conveyors (used to carry dry ingredients) regularly. Staff must be careful to clean the dead ends of conveyors where residues collect. They should clean the ends and joins between conveyors at least once a week. The free-flowing section of a conveyor should have hinged covers for easy cleaning and inspection. After conveyors are cleaned, workers should fumigate them with a fumigant that does not leave a residue. They should sweep and vacuum empty bins before fumigating them. Sanitarians should check material that is cleaned out of bins to make sure it is not infested. (Chap. 9 discusses pest control in dry storage areas.)

Brewing Area Sanitation

Spray cleaning is faster and works better than manual cleaning and can reduce the amount of time production has to stop for cleaning. Cold water can be used, but a water temperature up to 45°C (113°F) can help the cleaning compound react with the soil. In glass-lined tanks, the water should not be hotter than 28.5°C (83°F) because sudden temperature changes can break the glass. Temperatures hotter than 45°C (113°F) are not recommended because hot water causes condensation and products need more refrigeration. Management should install high-temperature cut-

off switches to control water temperature. Caustic-soda cleaning compounds attack soldered equipment and should not be used. Scale can be removed from aluminum vessels using 10% nitric acid, applied as a paste mixed with kaolin.

Hoses and fittings are expensive to buy and maintain. In the long run, stainless-steel lines are more cost-effective because they are so durable, even though stainless steel is quite expensive initially. Industrial spray nozzles can be put in places where they can clean vapor stacks on kettles, strainer troughs in hop strainers, and conveyor belts. Workers should clean the brewing area at least once a week and should remove debris and other obvious dirt every day.

Beerstone is mainly organic matter in a matrix of calcium oxalate and is one of the most difficult beverage soils to remove. A strong chelating agent, an alkaline cleaning compound, and heavy scrubbing work best.

Bottle Washing

Workers must carefully examine all empty bottles when they are returned by the retailer and the bottler. They should also inspect new bottles to make sure there are no obvious signs of contamination. All new and used bottles should be washed immediately before filling. A mechanical washer that sprays plenty of caustic solution on the inside and outside of the bottle works best. The temperature of the water used for spraying and rinsing should be 60 to 70°C (140–158°F). The final rinse can contain up to 0.5 part per million (ppm) of chlorine without affecting the flavor of the beer. But chlorination is necessary only if the rinse water is not pure.

Beer Pasteurization

Most brewers pasteurize their beer to make sure the condition, flavor, and smoothness do not change during distribution. Some brewers use sterile filtration instead of sterilization. If used, staff should replace filters every other week so that microbes cannot grow in the filters.

Beer is often pasteurized when it is bottled or canned because it is unlikely that the beer will be contaminated again after it is sealed. Overheating during pasteurization can spoil the flavor and make the beer hazy. So it is important to use the shortest amount of time and the lowest temperature possible while still destroying enough microbes. Most breweries now pasteurize using a conveyor system with a pasteurization cycle that takes about 45 minutes. During pasteurization, the beer is heated slowly from 1 to 2°C (34–36°F) to 61 to 63°C (142–145°F) and then cooled to room temperature. The speed of the conveyor belt controls how long the beer is heated.

Haze in beer may be nonbiological or biological. Nonbiological haze forms when products that are not very soluble settle out of the beer. This process is quicker when oxygen (air) is present. Biological haze forms when bacteria or yeasts grow in the beer. If the beer stays long enough in the cold conditioning tank and goes through fine filters it is unlikely to have nonbiological haze. Keeping air out of the beer container and choosing the right materials for containers also make nonbiological haze

unlikely. Other hazes can be caused by metal, especially tin. If beer has a haze due to bacterial or yeast growth, it was probably not filtered properly or was infected after filtration. A bacteria or yeast haze can be caused by poor sanitation in the plant or unclean storage containers or filters.

Cleaning Air-Conditioning Units

A good procedure for cleaning air-conditioning units follows:

1. Clean air-conditioning units every 6 months. Insert a ball spray through a special opening above the coils on top of the air-conditioning units.
2. Run water through the ball spray for 10 minutes to flush the unit.
3. Run a hypochlorite solution (200 ppm) at 40°C (104°F) through the ball spray for 5 minutes.
4. Let the unit soak for 5 minutes.
5. Rinse the unit with warm water for 10 minutes.
6. Check the unit and clean the pan bottom.

Conserving Water in Breweries

A wash-rinse cycle, sometimes called the *slop cycle,* can help breweries use less water during cleaning. This cycle is similar to the one used in most home dishwashers. A cleaning solution is pumped through a spray device for 20 seconds, left to soak for 1 minute for the cleaning chemical to work, and rinsed with water. The system is practical and economical because the cleaning solution is reused. The solution can be reused for longer if the tank that holds the cleaning solution has a top overflow to skim off floating soil and a drain valve at the bottom so that heavy soil can be drained out. The final rinse water can also be reused to prerinse another tank. This technique can reduce water and sewage treatment costs in areas that meter water use and sewage production.

WINERY SANITATION

Winemakers must remove any contaminants that spoil the taste, appearance, and keeping quality of wine. Several things can contaminate wine—for example, the reddish tartrate deposits that build up inside tanks during fermentation. It is important to clean processing equipment carefully to reduce growth of microbes everywhere in the winery. Wineries add sulfur dioxide (SO_2) at the end of the winemaking process to stop it from fermenting, but sanitary wineries only need to add very small amounts.

Sanitation becomes more and more important during the winemaking process and is most important at bottling time, but even the vineyard tools and harvesting equipment need to be washed to remove dirt, soil, and leaves. Destemmers, crushers, grape-processing areas, and bulk storage areas need brushing with detergent and

water. Harvester heads, pipes, hoses, pumps, faucets, spigots, and anything else that touches the juice or wine should be cleaned according to the five cleaning steps discussed earlier in this chapter. The same steps are used in the bottling line, but with extra control and checking to control the number of microbes in the wine.

The water used in a winery must meet chemical and microbial standards. Water can carry molds, yeasts, and acetic or lactic acid bacteria, so it is important to use pure water.

The design and layout of the winery should support good hygiene. Floors must be easy to clean with a nonslip, sloped surface. Walls and ceilings should be sealed and easy to clean. Winemakers should be careful to place equipment so that it does not create corners and crevices that are difficult to clean and so that floors are easy to clean. The equipment should have the same sanitary construction features as in other food-processing plants.

Cleaning Floors and Walls

Operation of a winery may be seasonal, but it needs to be sanitized year-round. Wet and dry cleaning methods are usually used together, depending on the type of soil. A heavy-duty, wet-dry vacuum cleaner can be used for floors, which should be cleaned at least once a week. Floors should be made of concrete and sloped, and should contain trench drains. Spilled wine, especially spoiled wine, should be washed away immediately. The area of the spill should be mechanically scrubbed, washed with lime or a strong hypochlorite solution, and rinsed with water. Floors should be periodically washed and disinfected with a dilute hypochlorite solution. Dry cleaning helps to keep the humidity of the winery low, which reduces mold growth. Tank tops, overhead platforms, and ramps can be vacuumed, cleaned, or washed, making sure that no water gets into the wine. The walls should be washed with a warm alkaline solution (such as a strong mixture of soda ash and caustic soda), rinsed with water, and sprayed with a hypochlorite solution containing 500 mg/L (0.67 oz/gal) of available chlorine.

Equipment Cleaning

Improperly cleaned equipment is likely to contaminate the wine. Crushers, pumps, lines, presses, filters, hoses, pipes, and tank cars are all difficult to clean completely.

Equipment should be:

- Taken apart as much as possible
- Thoroughly washed with water and a phosphate or carbonate cleaner for nonmetallic surfaces, and caustic soda or equivalent for cleaning metal equipment
- Sanitized with hypochlorite or an iodophor if the material being cleaned is adversely affected by chlorine

Enzymes are useful as cleaning agents because they can break down proteins, fats, and pectins (found on fruit). Enzymes are used in enology (winemaking) be-

cause they work best when the pH is near neutral (not acid or alkaline). Whenever possible, circulating the cleaning solutions is recommended. After cleaning and rinsing, hoses should be placed in sloping racks, rather than on the floor, to help them drain and dry. Equipment that has touched spoiled or contaminated wine must be thoroughly cleaned and sanitized.

During the harvest season, conveyors, crushers, and must (juice) lines should be kept clean. They should not stand with must in them for more than 2 hours. They should be washed, drained, and thoroughly flushed with water every 2 days.

Bottling-Area Cleaning

Good cleaning of the bottling area reduces contamination by bacteria and metals. Public-health agencies usually watch this area closely. The room should have good lighting and ventilation, glazed tile walls, and epoxy finished floors. Equipment should be far enough apart to allow cleaning in between, and the equipment should be easy to take apart. All pumps, pipes, and pasteurizers should be made of stainless steel so that debris, mold spores, and yeasts from freshly cut corks can be removed. Corks should be cleaned and sterilized before use by soaking for 2 hours in 1% sulfur dioxide and a little glycerol; and then rinsed with water. Corks can be sterilized using gamma-radiation to stop mold growth from causing off-odors.

Pomace Disposal

It is essential to get rid of the pomace (dry grape skins after the juice is squeezed out) as soon as possible after the grapes are pressed. Pomace must not stand in or near the fermentation room because it has a high acetic acid content, and fruit flies carry acetic acid bacteria from the pomace pile to clean fermenting vats. Pomace should be processed or scattered as a thin layer on fields. This thin layer dries quickly, so it does not become a serious breeding ground for fruit flies.

Cleaning Used Barrels

Alkaline solutions work best for removing tannins from new barrels. Barrels may be soaked in 1% sodium carbonate. If they need more cleaning, two atmospheres pressure of heated steam and several rinsings can be used.

In the past, empty barrels were washed with water and sprayed with a hot (50°C, 122°F) 20% solution of a mixture of 90% soda ash and 10% caustic soda. Then they were washed with hot (50°C, 122°F) water, sprayed with a chlorine sanitizer solution (containing 400 ppm of available chlorine), rinsed, drained, and dried using a dry-wet vacuum.

Any mold should be scraped off; because washing does not remove it. Washing with a quaternary ammonium compound and paints containing copper-8-quinolate help control mold growth. Burning a sulfur wick in the tank or the equivalent amount of sulfur dioxide from a cylinder of gas also works well. Before use, the tanks should be rewashed, and the barrels should be inspected and smelled before being

filled. A warm 5% soda-ash solution neutralizes vinegar in used barrels. This treatment should be followed by wet steaming and washing. If the soda-ash concentration is too high and the barrel is exposed for too long, it can damage the wood. The outside of wooden containers should be washed with a dilute solution once a year. Propylene glycol (400 ppm or less) can be used to control mold growth on tanks.

Removing tartrate deposits. Tartrate deposits make the inside surface of barrels very rough. Scraping is labor intensive and may injure the wood. A circular spray head inside the tank can help remove tartrates. Soaking with 1 kg of soda ash and caustic soda in 100 L of water also helps remove tartrates.

Storing Empty Containers

Concrete tanks should be left open and kept dry when not in use. They should be inspected and cleaned before they are used again. Open wooden fermenters are sometimes painted with a lime paste when they are not being used, but this is difficult to remove. It is better to clean the fermenters with an alkaline solution and then a chlorine solution. They can then be filled with water and about 1.6 kg of unslaked lime per 1,000 L (260 gal) of water. Stored empty barrels can be sulfured using a sulfur wick or SO_2.

Other Cleaning and Sanitation

Fillers, bottling lines, and other packaging equipment can be cleaned with CIP systems. A chlorinated alkaline cleaning compound can clean, sanitize, and deodorize at the same time if the soil is light. But the chlorine sanitizer cannot work if the equipment still contains organic matter.

Heat is the safest sterilization method. It can be used for everyday wines, but it is not good for high-quality premium wines because heat can change the taste. Sterilization in the bottle works best for sparkling wine.

Sterile Filtration

Filtration can sterilize wine via sterile filter pads or membranes. Diatomaceous earth filtration can reduce the yeast population but does not remove bacteria. This filtration can be followed by membrane filtration.

Reinfection

Most problems are caused by improper sterile filtration. Sterile filtration is ruined if the whole bottling line is not sterile. The best way to sterilize the whole system is to use a steam generator hooked to the filter and bottling line. A slow flow of low-pressure steam should run through the whole system for 30 minutes. The equipment is cooled using cold water before the wine is filtered and bottled.

Steam is not always available, and it can damage some equipment, such as plastic filter plates. Citric acid (300 g/hl [0.4 oz/gal] and 10 g/hl [0.013 oz/gal] SO_2 or

20 g/hl [0.027 oz/gal] metabisulfite) at a temperature of 60°C (140°F) is an alternative way to sterilize the system.

Some parts of the bottling line, such as the corker, are more difficult to sterilize. The corker jaws or diaphragm should be sterilized with alcohol. Membrane filters may be sterilized with water at 90°C (194°F).

Corks

Modern cork suppliers provide sterile corks. If winemakers are uncertain whether they are sterile, they should dip them in a 10 g/hl of SO_2 [0.013 oz/gal] solution before use.

Bottles

Bottles normally come in cases and can be contaminated by dust and cardboard. Bottles must be rinsed and sterilized using a solution of 500 ppm SO_2. Excess SO_2 is washed off the bottle using sterile water. An SO_2 dispenser can be placed on the main water supply.

Control

Sanitation standards must be checked during bottling. Winemakers can get special kits to check sanitation by counting the number of live yeasts or spoilage bacteria left in the wine after filling.

Pest Control

Fruit flies are attracted to fermenting musts. Many flies come to the winery from the vineyard. To control flies, crush grapes promptly after picking, remove all dropped and damaged fruit from the winery, get rid of all organic waste, use insecticides around the vineyard, wash all containers and trucks after handling grapes, and use insecticides on dumps. Flies are most active between 23.5 to 27.0°C (74–81°F), in dim light, and with little wind. Fans to blow air out of winery entrances, mesh screens, and air curtains help keep flies away.

Insecticides can kill fruit flies, but they only work on the area sprayed, and nearby areas may still have large fruit fly populations. If insecticides are used, they must be used at levels below the U.S. Food and Drug Administration tolerances. (Chap. 9 provides additional information related to fly, rodent, and bird control.)

DISTILLERY SANITATION

Sanitation Notes

As in breweries and wineries, the microorganisms that cause most problems in food-processing plants rarely cause problems in distilleries because of the ingredients,

processing techniques, and high alcohol concentration. However, distilled beverages can be contaminated. It is important to control the raw materials, because a contaminated finished product cannot be detoxified. When conditions in distilleries are not sanitary, the yield and the quality are poor.

Reducing Physical Contamination

Corn and other grains are inspected when they arrive at the plant. The biggest concern is insects, because a contaminated grain shipment can infect the storage silos and the entire plant. The most common insect pests in grain are flour beetles and weevils. It is also important to check grain for off-odors, because these can carry over into the final product. Grain storage silos are emptied 2 to 4 times a year, sprayed with high-pressure hoses, and allowed to air-dry. The area around the silos is kept clean by washing the area with water and spraying with insecticide.

Grain enters the plant on conveyors and should be sifted on shaking screens to remove corn cobs, debris, or insects. The mill room should be washed down with water to reduce grain dust, and it should be heated to 55°C (131°F) for 1/2 hour every 2 to 6 months to kill insects.

Reducing Microbial Contamination

Bacteria and wild yeasts can contaminate whiskey during fermentation. Most of the contamination comes from the malted barley. The malt is added at 60 to 63°C (140–145°F), so many of the microbes survive and grow during fermentation. Bacteria in the corn and other grains that are added before the cooking process or at temperatures greater than 88°C (190°F) are not a problem, because they are killed by the high temperatures.

Many microbes cannot grow during the fermentation process. At the beginning, sugar concentrations may be above 16%, and the pH is between 5.0 and 5.4 (acidic) for a sour mash whiskey. At the end the pH is between 4.0 and 4.5 (more acidic). Final alcohol concentration is about 9%, and the mixture contains plenty of carbon dioxide but very little oxygen. These conditions restrict the types of organisms that can grow.

To make sure the beverage is not contaminated during fermentation, dust is kept down by washing all plant surfaces (walls, floors, etc.) with water, and incoming shipments of malt are tested to check the level of bacteria. (Most distillers limit total bacterial counts to between 200,000 and 1,000,000/g.)

Equipment Cleaning

Large fermentation vessels (114,000–171,000 L, 30,000–45,000 gal) are cylindrical tanks with cooling coils on the inside. They should be cleaned by filling with hot water, detergent, and steam using a CIP system. After 30 minutes, the tank should be emptied, rinsed with water, and steamed for 2 to 3 hours to sterilize before new mash (crushed malt or grain) is pumped into the vessel.

The cooling coils should keep fermentation temperatures below 32°C (90°F). At hotter temperatures, the yeasts that cause fermentation die, and off-flavors develop. Cooling coils may collect "beerstone," which is hard and rocklike and contains calcium carbonates, phosphates, and sometimes sulfates. When beerstone builds up on the coils, they become less efficient at cooling. The beerstone should be removed every 6 months by filling the vessels with a 1% caustic solution (sodium hydroxide) and soaking for 3 days.

Residual grain builds up in the cookers, where the mash is prepared, and in the beer still, where the finished beer is pumped. The cookers, the beer still, and all connecting lines should be washed weekly with a 1% caustic solution. Some distillers use a 1% solution of acetic acid instead. The caustic solution can be prepared in a large tank and pumped to the beer still through the connecting lines and into each of the cookers. These areas should be rinsed with water to wash away all caustic residues.

The lines and stainless-steel tanks that carry distilled alcohol to receiving tanks and then into maturation barrels for maturation should be rinsed periodically. Because the product is crystal clear alcohol at 140 proof, sanitation does not need to be stricter.

Distillery products must be made using good-quality water. Water used to blend distilled spirits usually comes from a chlorinated and carbon-treated well or city water supply and is filtered. Chlorination makes the water microbiologically safe. Filtering makes sure the water is completely clear (no cloudiness).

SUMMARY

- Most soils in beverage plants contain a lot of sugar, dissolve in water, and are easy to remove.
- Microorganisms cannot be removed from the environment, so good sanitation is essential to control them.
- Strict control of raw materials is essential, because a beverage producer cannot detoxify a contaminated finished product.
- The bacteria that cause most problems in breweries are non-spore-formers. The best way to make sure beverage products do not spoil is to control contamination through a complete cleaning and sanitizing program.
- Spray cleaning works best with a properly blended, low-foaming cleaning compound, designed for the type of soil in the plant.
- Sanitizers—such as chlorine, iodine, or an acid-anionic surfactant—are recommended with the final rinse in fermenters, cold wort lines, and coolers.
- Sanitation becomes more and more important during the winemaking process and is most important at bottling time. Wet and dry cleaning are used together.
- Winemaking equipment should be taken apart as much as possible, thoroughly washed with water and a phosphate or carbonate cleaner for nonmetallic surfaces and caustic soda or equivalent for cleaning metal equipment, and sanitized with hypochlorite or an iodophor.
- Ways to remove tartrates include installing a circular spray head inside a tank or soaking with soda ash and caustic soda.

- Fillers, bottling lines, and other packaging equipment can be cleaned with a CIP system.
- Prompt processing of grapes after picking reduces fly infestation.

BIBLIOGRAPHY

Amerine, M. A., and Joslyn, M. A. 1970. *Table Wines: The Technology of Their Production.* 2d ed. University of California Press, Berkeley.

Amerine, M. A., Berg, H. W., Kunkee, R. E., Ough, C. S., Singleton, V. L., and Webb, A. D. 1980. *The Technology of Winemaking,* 4th ed. AVI Publishing Company, Westport, Conn.

Arnett, A. T. 1992. Distillery sanitation. Unpublished information.

Connolly, B. J. 1971. Stainless steel for wine storage and treatment installations. *Rev. Vin. Intern.* 92(180): 49.

Hough, J. S., Briggs, D. E., and Stevens, R. 1971. *Malting and Brewing Science.* Chapman & Hall, London.

Jefferey, E. J. 1956. *Brewing Theory and Practice,* 3d ed. Nichols Kaye, London.

Marriott, N. G. 1994. *Principles of Food Sanitation,* 3d ed. Chapman & Hall, New York.

O'Sullivan, T. 1992. High quality utilities in the food and beverage industry. *Dairy Food Sanit.* 12:216.

Remus, C. A. 1991. Just what is being sanitized? *Beverage World* (Mar.):80.

Remus, C. A. 1991b. When a high level of sanitation counts. *Beverage World* (Mar.):63.

Remus, C. A. 1989. Arrhenius' legacy. *Beverage World* (Apr. 1991):76.

Stanton, J. H. 1971. Sanitation techniques for the brewhouse, cellar and bottleshop. *Tech. Q. Master Brew. Assoc. Am.* 8:148.

STUDY QUESTIONS

1. Why are soils in beverage plants easy to remove?
2. Why is it important to keep beverages sanitary throughout processing?
3. What are the five standard steps for cleaning a beverage plant?
4. How do beverage producers purify water?
5. List one advantage and one disadvantage of sanitizing using heat.
6. Which part of beverage production needs the strictest sanitation?
7. What characteristics protect beverages from microbes?

TO FIND OUT MORE ABOUT BEVERAGE PLANT SANITATION

1. Contact a local winery, brewery, or distillery, and arrange a tour. Ask about their sanitation procedures.
2. Contact a soft-drink manufacturer (many have toll-free telephone numbers listed on the can or bottle). Ask where they obtain the water used in their soft drinks and how the water is treated.



GLASS OR PLASTIC BOTTLES FOR MINERAL WATER?

By law, natural mineral water should contain only the types of bacteria it contains at the spring. It is very difficult to keep other bacteria out of the water, especially when the water sits in large tanks and reservoirs before it is bottled. Bottling also affects the number and type of bacteria in the water, because the open system (open to the air) becomes a closed system (without air). The type of bottle (glass or plastic) also affects the bacteria. In glass bottles, the number of bacteria in the mineral water goes down after bottling. This is probably because residues of cleaning detergents in the bottles inhibit growth of bacteria. Organic substances that are dissolved in the plastic bottles have very little effect on bacteria in the water. Beverage producers bottling natural spring water need to be very careful to make sure that water is not contaminated with pathogenic bacteria and that these cannot grow in the water after it is bottled.

Source: Bischofberger, T., Cha, S. K., Schmitt, R., König, B., and Schmidt-Lorenz, W. 1990. The bacterial flora of non-carbonated, natural mineral water from the springs to reservoir and glass and plastic bottles. *Int. J. Food Microbiol.* 11: 51–72.